

Sponsored Project Checklist

It is the responsibility of the principal investigator to complete the form and to arrange for the required signatures from your Department Chair or School Director and the Dean. The PI must also provide a scanned copy of the signed form to the appropriate research admin office and contact.

Application Submission Deadline: 2026-10-01

Application Type: New Grant

Principal Investigator Information

Principal Investigator Information	Response
McMaster Investigator:	Hector Antonio Paez
Department/Institute:	School of Earth, Environment and Society
Email:	paezha@mcmaster.ca
Co-Investigators:	Lea Ravensbergen, Christopher D. Higgins, Moataz Mohamed, Juanita De Barros

Sponsor Information

If the funding is coming from a source other than the Primary Sponsor (e.g., by way of a sub-grant or sub- contract), please provide the name of the institution that is forwarding the funds. If it is not, please put N/A:

Sponsor Information	Response
Primary Sponsor's Name:	SSHRC - Insight
Contact Name:	SSHRC
Email:	webgrant@sshrc-crsh.gc.ca
Sponsor Type:	Government
McMaster Advisor is:	Amanda Graveline

Project Information

Project Information	Response
Project Title:	EquiTransit: Developing knowledge and tools to support equitable transit planning
Project Type:	New
If your project is continuing, please input the MOSAIC Project Number	No
Estimate Start Date:	2027-05-01
Estimate End Date:	2032-04-30

Sponsor Information A	Response
Sponsor Name:	SSHRC - Insight
Direct Research Costs:	\$500,000
Faculty Supervision Fees (if applicable):	\$0
Applicable Overhead:	\$0
Overhead Percentage:	0%
Total A:	\$500,000

This project has another sponsor: No

Sponsor Information B	Response
Sponsor Name B:	
Direct Research Costs B:	\$0
Faculty Supervision Fees (if applicable) B:	\$0
Applicable Overhead B:	\$0
Overhead Percentage B:	0%
Total B:	\$0

Research Security

Research Security	Response
Do any of the project sponsors include CIHR, NSERC, SSHRC or CFI?	Yes
Does the project aim to advance a Sensitive Technology Research Area (1) as defined by the Government of Canada's Policy on Sensitive Technology Research and Affiliations of Concern (STRAC Policy) (2)	No
If yes, as required by the STRAC Policy, have all researchers with named roles in the grant application (e.g. applicant, co-applicant, collaborator) completed the required form (1) attesting that they do not hold an active	

affiliation or receive funding or in-kind support from any of the listed Named Research Organizations (2)?	
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Certifications/Approvals

Certifications/Approvals	Response
Will the project create any safety hazards which are not addressed by protocols and Standard Operating Procedures that your group is currently using?	No
If yes, describe:	
Does this require the use of biological materials?	No
Does the project involve use of Humans, Animals, Biohazardous Materials or Controlled Goods as follows? a) human participants, their records or tissues ; b) animals and their tissues ; c) biohazardous materials (e.g. viruses, bacteria or yeast, cancer or immortalized cell lines, parasites, toxins of a biological origin, plant or aquatic pathogens); d) nuclear substances and radiation devices; or e) controlled goods (e.g. weapons, ammunition, explosives, weapon design and testing equipment, missile technology, technology necessary for the development, production or use of a controlled good)	No
Human Ethics:	
REB #:	
Animal Ethics:	
AUP #:	
Expiry Date:	
Biohazards:	
Expiry Date:	
Health Physics:	
Expiry Date:	
Controlled Goods and/or Technology:	
Expiry Date:	
Clinical Trial (drug trial involving human subjects):	
If yes, effective January 1, 2021 McMaster University is required to report on new Health Canada regulated clinical trials: Please complete the McMaster Clinical Trial Questionnaire.	
Does the project require an Environmental Assessment? (i.e. Does any of the research a) take place outside an office or laboratory, or b) involve construction, operation, modification, decommissioning, abandonment or other activity in relation to a permanent physical structure? For	No

additional information, visit the Canadian Environmental Assessment Agency website at <a):http:="" default.asp?lang="En&n=B053F859-1.):</a" href="http://www.ceaa.gc.ca/default.asp?lang=En&n=B053F859-1." www.ceaa.gc.ca="">	
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University Commitments and Facilities

University Commitments	Response
Does this proposal contain any financial commitment from McMaster University?	No
Source A:	
Amount A:	
Describe the nature of the commitment A:	
I have a second financial commitment source:	
Source B:	
Amount B:	
Describe the nature of commitment B:	

University Facilities	Response
Will you need additional space?	No
Location of additional space requested:	
Space Provider Name and Title:	
Will this project require any modifications to space?	No
If yes, please describe:	
Will you need to access to specialized facilities (e.g., Central Animal Facility, Faculty-specific Centres or Institutes)?	No
If yes, please specify:	
Have you arranged access with the facility director?	
Will any employees or researchers of the Sponsor be using the University's facilities in the conduct of the Project?	No
For the purpose of overhead distribution, is the work being conducted primarily in the Principal Investigator's home department?	Yes
If no, the primary facility / research centre is:	

Conflict of Interest

Do you, your co-investigator(s) or any member of the research team have any affiliation or a commercial or contractual interest with or in any of the Sponsor(s), suppliers or any other company associated with the project?:

Potential Conflict of Interest	Response
Do you, your co-investigator(s) or any member of the research team have any affiliation or a commercial or contractual interest with or in any of the Sponsor(s), suppliers or any other company associated with the project?	No
Principal Investigator Affiliation(s) or Conflict(s) of Interest:	
If applicable, please describe any received non-research compensation (cash or in-kind, including gifts of more than \$25) in past 3 years:	
Co-Investigator(s) Affiliation(s) or Conflict(s) of Interest:	
If applicable, please describe any received non-research compensation (cash or in-kind, including gifts of more than \$25) in past 3 years:	
Student(s)/PDF(s) Affiliation(s) or Conflict(s) of Interest.	
If applicable, please describe any received non-research compensation (cash or in-kind, including gifts of more than \$25) in past 3 years:	
Will the funding for this project originate from an agency covered by the Financial Conflict of Interest regulations of the U.S. Public Health Service?	No

Intellectual Property, Publication and Liability Issues

Potential Intellectual Property, Publication and Liability Issues	Response
Will graduate students be involved in the Project?	Yes
Is it part of their academic activity towards their degree?	Yes

Will this project generate intellectual property (IP)?	No
If yes, who will own the IP?	
Are you conducting any research for another Sponsor that might overlap with this Project?	No
If yes, please describe other Sponsor and overlap:	
Will the project be leveraged with other funding?	No
If yes – please select the funding agency(ies) and/or program(s):	

Signatures

Principal Investigator

I, Hector Antonio Paez, attest that all of the statements and answers are true to the best of my knowledge.	Hector Antonio Paez
Date	September 23, 2026

Approval Instructions

Department Chairs: Kindly reply “approve” to an automatic email send by sci_rsch@mcmaster.ca to indicate your approval status for this form. If you need additional details or verification. Please contact the PI directly

Summary (max. 3,800 characters)

Transit systems are a key component of urban public policy. These systems intersect with broader transportation and urban land use issues and are therefore of great interest and practical import. In general terms, transit systems generate accessibility to opportunities that contribute to the economic operation of urban regions. But this is not the only service that transit systems provide, since they use space more efficiently than low-capacity vehicles (mainly private automobiles), and thus help to reduce congestion and result in lower emissions per capita. Furthermore, transit is an obligated form of transport for many members of the public who may not be able to own or use a private vehicle and increasingly supports many urban residents who choose not to own a vehicle. As such, transit systems also help to reduce transportation-related social exclusion, and support democratic participation in many essential activities.

Despite their importance in public policy, transit services are often offered without explicit mention of the principles that underpin their provision. Implicitly, transit is often seen as an economic service that must remain fiscally responsible. But while fiscal responsibility is an important consideration, it is not by any means the only one. In reality, many of the benefits of transit systems are non-monetary, and a tendency to see them mainly as economic services leads to a narrow approach that can blunt their attractiveness. Particularly, there are numerous examples where this leads to major inequities in terms of the distribution of the benefits and the costs of public transportation, while ignoring its role as a facilitator of democratic participation in society.

Planning for transit systems that are equitable and promote ideals of transportation equity and justice is an urgent task, especially as the limits of highly individualized transportation systems become clear. EquiTransit proposes to re-envision transit from a broader, more just perspective, and aims to create the knowledge and tools needed to realize this vision. This project follows on the heels of recent research initiatives in Canada, particularly the SSHRC-funded project Mobilizing Justice, which have broken new ground and made considerable progress towards more inclusive, equitable, and just transportation systems in Canada. EquiTransit builds on these foundations with a more targeted focus on public transportation.

Accordingly, the objective of EquiTransit is to generate novel, theoretically-sound and analytics-based equity indicators useful to assess the performance of public transit systems in Canada. Theoretically, we will use a framework of justice that emphasizes the 5WH approach (why, what, who, where, when, how). To ensure that our indicators are pertinent to the question of "why equity?", we propose a Codesign approach, whereby the research team will work with transit operators and users in the explicit identification of the rationale and motivation for the development and use of indicators. We plan to use cutting-edge geospatial technologies to investigate the questions "who seeks equity, and where and when?". Further, we propose to create open-source tools, including equity-informed planning and assessment tools, and place them at the fingertips of stakeholders at the lowest possible cost. Finally, in the spirit of Codesign, the effectiveness of the tools will be assessed in consultation with stakeholders, so that refinements can be incorporated in successive releases of the tools. In this way, EquiTransit aims to produce actionable knowledge to support the development of more equitable transit in Canada, to the benefit of urban residents, democratic governance, and democratic participation in Canadian society.

Detailed Description (max. 6 pages)

Objectives: Generate novel theoretically-sound and analytics-based equity indicators to assess the performance of public transit systems in Canada. Investigate the reliability of the indicators by studying the experience of transit users in areas with equity gaps. Create open-source tools, including dashboards, to make these indicators available to stakeholders, such as transit practitioners, policymakers, residents, and advocates. Assess the effectiveness and accessibility of the tools in consultation with stakeholders and incorporate feedback in newer releases of the tools.

Rationale: Traditional public transport performance evaluations usually rely on objective ^[OBJ:OBJ], such as operational efficiency^[OBJ:OBJ], reliability^[OBJ:OBJ], safety^[OBJ:OBJ], and objective comfort^[OBJ:OBJ], often leaving aside equity considerations related to both the spatial and temporal coverage of transit service(6)^[OBJ], and consequently failing to meet the needs of the most disadvantaged residents(7)^[OBJ]. The transit planning process primary focuses on service cost (8–10)^[OBJ], operational features(11–13)^[OBJ], and emissions^[OBJ:OBJ]. Furthermore, current standards (e.g., Transit Capacity and Quality of Service Manual (TCQSM)) and manuals (e.g., Federal Transit Administration (FTA)) completely overlook equity dimensions in their transit service planning, operation, and evaluation.

Yet equity-deserving groups are those who rely most on public transit (17–20) and, when service is insufficient (21,22), face the greatest risk of social isolation due to transport-related factors, preventing them from participating in activities that are essential for the economic and psychological wellbeing (8,23,24). This group is often characterized as Captive-Users, as their mobility choices and activity participation are limited to transit level of service (LoS), both spatially and temporally.

Approaches: The project aims to produce actionable knowledge to support the development of more equitable transit in Canada. At one level, we will produce equity indicators firmly grounded in sound theories of justice. We consider this aspect of the project the one of greatest academic interest. Further, to ensure that the indicators are not only theoretically sound, but also of practical relevance, we will use them as a framework to research the experience of equity-seeking transit users in judiciously selected areas. A key aspect of procedural justice is ensuring that all stakeholders participate in the planning process and have access to the information they need for decision-making. For this reason, we will adopt a codesign framework to design equity indicators in partnership with stakeholders in Hamilton, Ontario. Implementation of the indicators will use General Transit Feeds Specifications (GTFS), the standard for evaluation and planning of transit operations, and will implement an equity analytical module to complement the usual cost-based evaluation of transit systems. To facilitate the exploration of indicators and the evaluation and planning with the equity evaluation module, we will develop open-source tools (EquiTransit) that will embed all elements in an accessible interface (a dashboard) that will code best practices in data-analytics. The tool will be based on open data and will be able to use data from different regions. Once the tools have been tested within the codesign framework in Hamilton, we plan to implement the tools for Canada's 14 largest Census Metropolitan Areas (CMAs). The aim of the tools is to allow for the examination of socioeconomic and spatial-temporal inequalities in public transportation services, based on geo-spatial technologies. We also will assess the efficacy of the tool by investigating the experience of users. This information will help us to refine the clarity of the information, and possibly to adapt the tool to the levels of data literacy of the populations most likely to use it.

Anticipated Benefits: The impacts of inequitable access to public transit are thought to be large, particularly for equity-deserving populations, yet spatial-temporal inequalities in transit provision remain little understood and of peripheral interest in transit planning. Clear and transparent tools will allow these considerations to be integrated into planning practice. Codesigning the indicators and tools means that their legitimacy is increased, since the process of design is not top-down, and can empower policy makers to make bolder decisions with greater public support. More concretely, including equity indicators in transit performance assessments can help uncover inequalities, enabling planners and advocates to understand how disadvantaged populations in different locations and time periods are, or are not, served by public transportation. For example, visualizations showing the frequency of public transit service across areas of different income levels and at different times of day can help practitioners identify where and when low-income populations are underserved, highlighting where interventions are most needed. Furthermore, performance metrics will be able to be disaggregated by income, race, gender, and other sociodemographic characteristics, in addition to their traditional spatial and temporal dimensions. In this way, they can be examined by residents and advocates to identify populations and neighbourhood with insufficient transit

service. This increases awareness of low spatial-temporal coverage for disadvantaged populations and helps communities pressure policymakers to intervene, supporting their right to access transportation.

Key Challenges: Synthesizing available knowledge on the equity distribution of public transit service into meaningful and actionable information for planners, residents, and advocates. This will be accomplished by combining expertise in transport equity and justice, codesign, public transportation, qualitative research, transport data analysis, and spatial data science. Codesign and stakeholder engagement will ensure that the tools produced are meaningful and useful in practice.

Facilitation of knowledge transition: This project aims to develop the EquiTransit tool for community planners and members to better identify who has the most and least amounts of different transportation services available, but even more crucially, where those populations are located and when are the services needed. Using spatial and temporal visualizations, planners and other stakeholders will be better able to address issues in equity in transportation access, especially via public transit and walking to reach transit services in any hotspots with fewer transportation options available. This would be particularly beneficial for transit agencies to examine to better understand where service changes or additions should be made to increase transit accessibility within given walking distances and potential travel destinations. Additionally, the associated knowledge and findings derived would serve as a framework for other academic researchers to further develop their own findings, allowing for more research into transit inequality based on different aforementioned sociodemographic factors such as race, gender, income, education level, among other characteristics.

Public Transportation and Equity: Traditional transit performance evaluations optimize service for an average user by focusing on efficiency, ridership, and reliability. These are system level, operational performance metrics that frame transit evaluation in a way that does not grapple with social outcomes. Yet, transit is most heavily relied upon by individuals who have the fewest alternatives, namely captive users, which makes the distribution of service to different people, at diverse places, and various times a question of justice (25). Distributive justice in transportation asks who benefits from the system and who carries its burdens, shifting the evaluative criterion from system performance to the accessibility opportunities delivered to different socioeconomic groups (26). This framework moves beyond horizontal equity, which treats all riders identically, to vertical equity, that is, actively directing better service, coverage, and frequency to the groups who rely on it most and face the greatest systemic barriers. Vertical equity, it bears noting, represents a stricter idea of justice, as new benefits are unequally distributed in favor of under-privileged populations.

Evidence from Canadian sources supports this concern. In 2021, 17.6% of racialized commuters in large urban centres took public transit to work, compared with 6.4% of non-racialized commuters, and 13.1% of immigrants relied on transit versus 4.7% of those born in Canada (27). By May 2025, roughly one in four Black (25.7%), Latin American (25.3%), and Filipino (24.4%) commuters used transit, and the average transit commute had reached 44.1 minutes, nearly twice that of a car commute, with most racialized groups facing systematically longer journeys (28). These patterns coincide with material deprivation. Allen & Farber (24) estimate that close to one million Canadians across the eight largest CMAs live in low-income households in areas of poor transit accessibility, placing them at risk of transport poverty. Women in lower-income households also undertake substantially more care-related travel that are poorly serviced by current network designs (30).

Existing planning tools rarely delve into these inequities. A recent analysis of Toronto's bus network shows that although two-thirds of TTC riders are equity-deserving, most live outside the agency's designated equity priority areas, leaving much of the population the framework is meant to serve largely invisible to it (20). Traditional performance metrics, and even most accessibility measures, also average across the day, masking systematic temporal under-service during evenings, nights, and weekends, when shift workers, who are disproportionately racialized, immigrant, and low-income, most depend on transit (17). EquiTransit will close these gaps with metrics and tools that bring distributive justice into transit planning across both space and time.

WORK PACKAGES

WP.1 Theoretically-grounded equity indicators: In the transportation context, it is well established that the burdens and benefits of transportation are distributed unevenly across space and time, with some groups receiving greater benefits at the expense of others (20). To better understand how these benefits and burdens are distributed

among a diverse population, a theoretical framework related to equity and justice is needed. In this proposal, we follow recent work of the PI with members of the SSHRC-funded project Mobilizing Justice team (26). This work adopted the framework proposed by Jaggard (27) and adapted it to the particular conditions of the transportation context. According to this framework, the examination of equity, fairness, and justice is guided by six interrelated questions: Why? What? Where? When? Who? and How?

Why? Justice is a broad moral and political concept that relates to: (1) the rights and entitlements that should be recognised and enforced; (2) the fairness of processes and procedures in decision-making and distribution (procedural justice); and (3) the appropriate distribution of benefits and burdens of social life in society. The last concept, also known as distributive justice, is perhaps the most studied form of justice in the transportation literature and has as its purpose to provide a fair redistribution to repair past harms (31).

Pereira and Karner (34) argue that future studies on distributive issues in transportation equity would benefit from justice theories that link two perspectives: Rawlsian egalitarianism (33) and the Capability Approach (35–38). Rawlsian egalitarianism is grounded in two interconnected principles, with the Principle of Greatest Equal Liberty as the first, stating that every person has an equal right to the most extensive system of basic liberties that is compatible with a similar liberty for everyone else. The author lists as basic liberties the freedom of thought and conscience, freedom of speech and assembly, political liberties, freedom of the person (from arbitrary arrest), and the right to hold personal property.

The second principle of Rawls (33), more specifically the Difference Principle, states that inequalities are only justifiable if they benefit the least advantaged members of society and derive from conditions of fair equality of opportunity. The first principle is hierarchically above the second. For Rawls (33), an unequal policy is only acceptable if this directly translates into an increase in benefit for the worst-off population, and if it does not disrespect other individuals' basic rights and liberties, which must be applied equally to everyone. For instance, even if a policy or project improves a certain transport-related benefit for an equity-deserving population, it is unacceptable if it infringes on others' basic rights and liberties. The Difference Principle is centred on the distribution of what Rawls named primary social goods: various social conditions and all-purpose means necessary for individuals to pursue their life plans, representing fundamental resources, rights, and opportunities that, regardless of their specific life goals, every rational person is assumed to want. They are called primary social goods because they result from human cooperation and not from a natural fact, and they are primary because it is necessary to have them to maintain the status of a free and equal citizen (34).

Sen (30–33) criticizes and builds upon Rawls' theory, arguing that the focus of the Difference Principle should change from primary social goods to human capabilities, defined as the collection of opportunities available to individuals and the freedom to choose and to act. This perspective is known as the Capability Approach. This approach recognises the diversity of human preferences and needs and the role of the environment in improving or constraining the set of opportunities available to individuals. Sen affirms that reducing inequalities in capabilities is important to promote equality of opportunities and guarantee minimum levels of basic capabilities.

What? In this research proposal, we argue that public transportation should be the focus of distributive justice. Public transportation is essential for promoting more sustainable cities due to its potential to reduce traffic congestion and carbon emissions and to improve air quality (35–37). In addition, public transportation increases people's accessibility, the main benefit of the transportation system (20), understood as the ease of reaching spatially distributed opportunities (38,39). In fact, public transportation provides access to opportunities for those who cannot - or prefer not to - drive, walk, or cycle.

Access to opportunities and destinations is the most auspicious indicator for distributive justice (20), the most comprehensive indicator of the performance of land use and transport systems (40), and one of the main outcomes of spatial development (46). Scholars argue that this measure serves as a comprehensive and transparent indicator of transportation supply, with the additional advantage of being replicable across different jurisdictions and periods (41). Accessibility is a required condition for increasing people's freedom of choice and promoting equality of opportunities (41). An adequate level of access to opportunities, which can be offered by well-planned public transportation system, can increase the chances of attending a doctor's appointment, watching a movie at a cinema, engaging in a sporting activity in a park, or finding and keeping a job.

Who? Disadvantaged population groups, including those affected by low income, limited ability, and historical discrimination, have become a key focus of transportation research (28,42). This is due to the potential of historical discriminatory policies and other disadvantages to negatively impact their transport experiences. Additionally, these groups have often been excluded from the transportation planning process, resulting in their needs frequently being not recognized. Relevant groups include youth, adults, women, children, low-income individuals, zero-vehicle households, people with disabilities, single-parent households, refugees, and racialized populations (43–48).

How? Both justice theories, Rawlsian egalitarianism and the Capability Approach, address sufficientarian and egalitarian moral concerns regarding how benefits and burdens should be distributed. While these two concepts are frequently applied in transportation equity research, they are often not explicitly articulated (19,49). The first approach, the sufficientarian perspective, focuses on why some groups or areas lack sufficient access to those goods and services. This approach builds on the idea of a ‘sufficient threshold’, that is, minimum levels of transport goods, services, and accessibility to essential activities that should be guaranteed to everyone to meet basic needs and lead to a meaningful life. In this idea, absolute priority should be given to improving the well-being of those whose well-being level falls below the sufficient threshold.

The egalitarian perspective frames fairness in terms of the relative distribution of goods and bads among members of society. Egalitarians focus on inequalities between social groups or geographical areas, asking why some groups and areas have higher or lower levels of access to transport goods and services. This perspective holds the premise that all people should be treated equally, focusing on relative differences, not on absolute inequality levels. For instance, it considers it unfair if a population group has relatively lower access due to discriminatory policies, even if their access level exceeds a minimum threshold (28).

In practice, transportation analyses generally rely on standards to determine whether a situation is fair (26). Once operationalized, an equity standard defines the level of fairness to be adopted and the acceptable level of inequality, usually combining a concept of fairness with threshold criteria, thereby reflecting both vertical and horizontal approaches to equity. Vertical equity is typically operationalized through opportunity-based standards defined for specific population groups (e.g., minimum access to opportunities within a 10-minute bus ride for older adults), whereas horizontal equity is generally operationalized through uniform travel barrier thresholds applied to all potential users (e.g., a maximum acceptable distance to a public transit stop). Regarding the definition of equity standards, our approach is guided by the principles of sufficiency and egalitarianism. This means that the minimum level of accessibility to public transportation should be defined according to the needs of the most equity-deserving population, thereby reducing inequalities in opportunities.

Where? By their very nature, interventions in the transportation sector are spatial interventions, and, due to the structuring effect of space (50), they inevitably generate inequality by benefitting those who live in or have access to the areas served by transportation infrastructure and services, while at the same time potentially harming others, either directly or indirectly (26). It is not uncommon for transportation policies to provide a sufficient or more than sufficient level of accessibility to the majority of people who own cars (51), while failing to ensure a basic level of accessibility for those who do not have access to a car, cannot use one, or do not wish to use one. To address this imbalance, improvements are needed in public transportation networks that provide users with a level of freedom comparable to that enjoyed by car users (52).

The spatial dimension of analyses of transportation equity is often based on the concept of spatial equity, according to which cities should seek to distribute the benefits of transportation systems equally across space (31). Although transportation equity requires spatially targeted policies and interventions (34), rather than broad approaches that encompass all of society, our perspective views equity primarily as an issue concerning people. Consequently, while we intend to identify areas with insufficient public transportation service and accessibility, these areas will be identified through measures and analyses focused on equity-deserving populations and facing insufficient access to public transportation, rather than solely on areas with low service. From this perspective, areas with a higher concentration of disadvantaged populations should take priority over wealthier areas, as disadvantaged groups tend to rely more on public transportation (28). Finally, our analysis will be limited to metropolitan areas and, within them, to urbanized areas, having the public transportation stops and stations at the fine-grained spatial level of analysis. We do not focus on rural areas, as rural residents may have different expectations and needs regarding public transportation services.

When? To ensure justice in transportation, we evaluate the temporal aspects of transportation interventions. For instance, we investigate when a specific intervention takes place, the historical policy context in which it occurs, and for how long its benefits and burdens can be attributed to that intervention. In addition, the temporal dimension is central to our analysis because we are interested in investigating how equity-deserving populations may receive insufficient public transportation service, identifying not only the areas but also the times of day when this insufficiency is most pronounced.

WP.2 Codesign: In the 5WH framework, the question of WHY? constitutes the cornerstone of our approach to analyzing equity. This question, however, is not answerable in a strictly technical fashion. Data can help to support it, but to produce truly actionable knowledge public transit users and practitioners must co-create the equity indicators developed to assess the performance of public transport systems in Canadian cities. For this reason, CODESIGN is an essential component of the project, and will run throughout the life-course of the project, from the development of indicators, the choice of standards, and the testing of open-source tools.

Codesign is a form of community-based participatory action research; it is a collaborative approach that brings together researchers and stakeholders to address the needs of the community, enable social system change, and increase the impact of research. Stakeholders are placed at the centre of the design process to effectively solve problems and improve human experiences (53–55). In this project, the stakeholders will include the research team, public transit providers, and users.

Though the tools for this overarching proposed research project will be developed for Canada's 14 largest CMAs, the Codesign process will focus on the City of Hamilton, where McMaster is located and where the strongest community-based research already partnerships exist. Research collaborators working at the HSR will be involved as the public transit providers. Public transit users will be selected from Hamilton's 'code red' neighbourhoods; areas that experience the highest rates of poverty, unemployment, and chronic illness in the city (56).

The research will follow the 5-step human centered Codesign approach (57). Empathy is the first step. We will begin by understanding service users' experiences with public transport systems as well as their understandings of the standards the currently exist. This will be done through a series of Codesign focus groups. The second and third steps, to Define the key issues, and to Ideate, i.e., to brainstorm possible solutions, will take place in a second round of focus groups. This time, a series of separate Codesign focus groups will take place with users and with transit providers. PhD student 1 will lead these three sets of focus groups and present the results to the larger research team. The larger team will then complete step 4, which is to Prototype a variety of tangible solutions based on the focus group discussions so far. The final step is to Test and re-design the solution as needed. Here, PhD student 1 will complete a final round of focus groups with users and providers in which they will present the prototyped solutions for feedback. These results will be presented to the research team again, with the possibility of another round of re-design and testing. The final outputs should include new tools that are not only theoretically and empirically sound, but that also respond to the needs of transit users and providers.

All Codesign focus group sessions will be supported by McMaster university's equity-based Co creation hub (eqCC). This hub unites researchers, students, service providers, and users to engage educate and innovate for systemic transformation (58). All focus groups will take place at McMaster University, and all participants will be compensated for their time with a \$50 gift cards per focus group they participate in. They will also be reimbursed for transportation (either reimbursed a bus fare or parking will be paid for).

WP.3 Geospatial data analytics: EquiTransit will be built entirely from open data and open-source software, ensuring transparency and reproducibility across jurisdictions. For each Census Metropolitan Area, we assemble four open inputs: sociodemographic data from the 2026 Canadian Census (via the {cancensus} interface to CensusMapper from von Bergman et al. (59), General Transit Feed Specification (GTFS) schedules published by transit agencies, OpenStreetMap road and pedestrian networks, and the Global Human Settlement Layer built-up surface. To enable consistent comparison, these inputs are harmonised onto a fine hexagonal grid; dasymetric interpolation, restricted to residential cells, apportion census population and attributes to each hexagon. Using the {r5r} routing engine (60) we then compute different measures of transportation accessibility. This includes walking travel times between populated hexagons and transit stops to estimate the population effectively served by each stop. Larger network accessibility analyses will also be conducted to capture how well the transit system enables

individuals to reach key destination types, such as places of work or daily necessities, around each study area within a reasonable travel time.

Critically, the travel time and accessibility analyses will be conducted using ten-minute transit service intervals, which is done to capture how level of service (e.g. service frequency) varies through the day and expose temporal gaps that daily averages conceal. GTFS Realtime data, which captures actual transit vehicle movements rather than static schedules, offers a particularly interesting avenue to better understand how service levels and accessibilities are provided. Building on initial schedule-based analyses, we will expand our lens to incorporate realtime data over sampled time windows as the project progresses, making use of new computational tools like the `{gtfsrealtime}` (61) package for R. From this, the transit accessibilities are disaggregated by income, race, gender, and immigration status, revealing inequities across both space and time. The complete workflow will be packaged as a reproducible analytical pipeline and an interactive dashboard, initially covering Canada's fourteen largest CMAs, with parameters, indicators, and presentation refined through stakeholder consultation.

WP.4 Equity-informed transit planning and evaluation: Transit planning is often planned on legacy routes and updated as population and urban geography expand. Then the service is packaged into a global universal standard: General Transit Feed Specification (GTFS). While the planning process itself is mathematically challenging, it is fundamentally developed under a cost-reduction framework, maximizing the utilization of resources, and balancing operation with political decision and budget constraints. The well-established four-step transit design model (e.g., routes, frequencies, fleet, and crew assignment) does not include any SED elements that capture the equity dimension of transit users. We will develop two interventions enabling a novel implementation of equity transit planning.

For existing network, a GTFS-based equity analytical model will be conceived, developed and deployed. The model will utilize equity indicator, geospatial data, and findings from the CODESIGN, and integrate them with GTFS files. Furthermore, Equity KPIs will be embedded in a GTFS-analytical model, enabling transit agencies, policymakers, and practitioners to evaluate the equity gaps in existing design.

For new service design, the equity indicators will be integrated in the four-step transit model, offering a novel equity-based transit planning framework transferable to any context. Equity-indicators will be integrated in all four steps of the planning process. For example, frequency estimation is based on passenger flow/demand, wait-time, and fleet size (e.g., available resources). Although passenger flow/demand is an indirect indicator of equity, it remains ineffective of differentiating service provision for equity-deserving groups compared to other users.

Together, the GTFS-equity analytical models and the equity-informed four-step transit planning models will be assessed using deployed to Canada's 14 largest CMAs. In itself, this validation process offers a substantial transit evaluation databank transferable to these CMAs.

Knowledge Mobilization Plan (max. 1 page)

The research team will implement audience-appropriate Knowledge Mobilization (KM) strategies. Key audiences are 1) members of the public; 2) transportation planners and practitioners; and 3) academic and scientific peers. The creation of transparent, research-based tools will improve planning processes and outcomes.

Public and practitioners: A key aspect of the proposed work is the use of a codesign framework. In codesign there is a continuous and multilateral flow of knowledge during every step of the project. In addition to the involvement of members of the public and practitioners in this endeavor, we will expand the diffusion of this research by writing general reports in layperson language to facilitate knowledge transfer. These reports will take the form of policy briefs, user's manuals, and sample case reports related to the release of the tools. These will be posted publicly via a website that will be hosted on GitHub Pages. Members of the research team have experience developing and releasing available materials for public use with this medium (62,63).

Academic: This research proposal includes 4 research students (2 Masters and 2 PhDs), two post-doctoral researchers, five professors, and one international collaborator. The goal is for each Masters student to lead at least one paper for a national conference and then publish in an international peer-reviewed journal. PhD students will aim to publish three peer-reviewed journal articles. The post-doctoral fellows will lead one research paper each. Thus, 10 papers would be led by trainees. Although ambitious, we have identified sufficient challenges and steps to the project to warrant this number. This project primarily has implications for planning in urban regions. Accordingly, the target journals for this research include *Transportation Research A: Policy and Practice*; *Public Transportation*; *Transport Policy*; *Journal of Transport Geography*; *Transportation*; the *Journal of Transport and Health*; and *Travel Behaviour & Society*. In addition to publishing Open Access, members of the team are on websites such as ResearchGate.net where pre-print versions can be made available. Further, the project will have a GitHub Pages website where updates will be made each academic session (every four months) and when significant events have occurred (e.g. presentations, publications, workshops, etc.). The project will also be promoted by the collaborators on websites such as "young-mobility.at/en". Conferences will range from local to international and will target audiences with both researchers and practitioners. To further help with knowledge diffusion, the researchers will propose workshops to relevant conferences such as the Annual Meeting of the Transportation Research Board in Washington, D.C. and the Annual Meeting of the Association of American Geographers. Members of the research team have experience conducting such workshops.

Open-Science Statement: The Tri-Council system in Canada (i.e., CIHR, NSERC, SSHRC) is currently phasing in an Open Access policy for the research it funds. There are important reasons for this, as calls to make data as open and reproducible as possible become increasingly urgent. But Open Access of publications and data is only one facet of open science. Frankenhuys and Nettle (66) argue that encouraging quality of work throughout the research process requires an effort not only to make research findings open, but the data-collection process, methodology, and analysis as well, thus dissuading scientists from simply focusing on obtaining 'ideal' findings. Additionally, open science encourages and increases engagement from members of the public regarding ongoing research and any relevant issues that can emerge (64). In this respect, members of the research team have already a proven record of creating open, reproducible science, not only by publishing preprints publicly and sharing data, but by sharing every aspect of research, from the stage of data preparation and up to reproducible computational notebooks. Examples of data sharing by members of the research team include the creation of several open-data products (65) such as ActiveCA package (68) and TTS2016R package (62). An example of equity-based research in transportation that illustrates the mode of sharing pre-prints along with reproducible notebooks is Desjardins et al. (66). Materials such as this, shared publicly on websites like GitHub, allow for thorough verification of the research, to ensure it does not contain major errors that could impact the validity of the results (67). Open-science practices like this also speed up the process of discovery, by making it easy for other researchers to access the base work and modify it to accomplish whatever is required for their own specific work. The members of the research team are experienced open-science researchers and are fully committed to following Tri-Agency policy on Open Access, and more broadly to continue producing high-quality research that is open, transparent, and reproducible.

Research team, previous output and student training (max. 4 pages)

A. DESCRIPTION OF RESEARCH TEAM

The theme of public transportation and equity is multifaceted and calls for a research team with an appropriate mix of skills and expertise. In response, this project gathers a group of highly qualified researchers with complementary skills and perspectives. The research team is composed of one principal investigator, four co-investigators and one international collaborator, and includes scholars in the social sciences (Páez, Higgins, Ravensbergen, and Pereira), humanities (De Barros), and engineering (Páez and Mohamed). The principal investigator (Páez) and two of the co-investigators (Higgins and Ravensbergen) are alumni of the SSHRC-funded project Mobilizing Justice. One of the co-investigators (De Barros) is Director of McMaster Centre for Human Rights and Restorative Justice, and another (Mohamed) is Director of McMaster Centre for Transportation and Logistics. The international collaborator (Pereira) is a highly regarded scholar whose work on transportation equity and the development of open software for transportation is peerless. The profiles of the members of the research team, along with their expertise and roles in the project are as follows.

Christopher D. HIGGINS (co-applicant, relative contribution to project 15%) is Associate Professor at the University of Toronto Scarborough in the Department of Human Geography and in the Department of Geography and Planning at the University of Toronto. He is among the world's top 2% of researchers in the Elsevier-Stanford rankings and received the 2025 University of Toronto Scarborough's New Scholar Award for exceptional research achievement and impact among early career faculty across disciplines. He has experience working with public sector partners such as the City of Toronto, City of Hamilton, and Ontario Ministry of Transportation. His research uses geospatial analysis and accessibility measurement to examine how transportation systems shape access to housing, employment, and essential services, with particular attention to social equity. His work has been supported by SSHRC and NSERC, most recently through an NSERC Alliance Grant supporting the University of Toronto's tri-campus Mobility Network Consortium. For EquiTransit, he will lead the development of geospatial analytics, accessibility indicators, reproducible data pipelines, and interactive analytical tools. He is committed to investing 15% of his research time in this project.

Juanita DE BARROS (co-applicant, relative contribution to project 15%) is Professor of History at McMaster University and the director the Centre for Human Rights and Restorative Justice. She is an internationally-recognized historian whose work explores the intersection of race, gender, and class in the construction of public policy, including in colonized urban spaces. This domain of knowledge is key to the project, as it combines social and government policy in ways that affect racialized and genderized people in urban spaces. She has published widely, including 2 monographs and over 20 scholarly articles and book chapters. She has also coedited 2 journal special issues, 3 scholarly edited collections, and co-written 5 introductions to edited collections and journal special issues. She is the co-editor of two scholarly book series, one with Cambridge University Press and other with McGill-Queens University Press (Confronting Atrocity: Human Rights and Restorative Justice). Dr. De Barros expertise in discourse analysis and in the field of human rights is central to the aims of this project, where a clear understanding of stakeholders about the grounds for justice in public services is essential. For this project, she will lend her expertise and vision through the cosupervision of a post-doctoral fellow and a doctoral fellow. **She is committed to devoting 15% of her research time to this project.**

Moataz MOHAMED (co-applicant, relative contribution to project 15%) is associate Professor of Smart Systems and Transportation at McMaster University and Director of the McMaster Institute for Transportation and Logistics (MITL). He is the 2025 recipient of the Joseph Ip Distinguished Engineering Fellowship and is featured Elsevier-Stanford list of the world's top 2% of scientists. Prof. Mohamad is an experienced researcher, with more than 90 peer-reviewed articles and over \$6 million in secured research funding to his credit. He has trained 57 highly qualified personnel, including 11 PhD students, 12 master's students, and 7 postdoctoral fellows. Since 2018, he has been principal investigator on a City of Hamilton-funded program to reenvision the city's transit operator, Hamilton Street Railway (HSR). He is the principal developer of the AI-Transit Orchestrator, an \$1.4M NRCan-funded fleet-planning toolkit co-developed with Halifax Transit, Calgary Transit, Niagara Region Transit, and ETUQ. On this proposal, he will lead the team's work to integrate equity considerations in transit planning operations, translating

multi-agency networks into a GTFS-based equity analytical model and equity-informed planning framework. **He is committed to allocating 15% of his research time to this project.**

Antonio PÁEZ (applicant, relative contribution to project 30%) is Professor of Transport Geography at McMaster University. He is a seasoned researcher who led the multi-institutional team that won the 2010 Meredith F. Burrill Award of the American Association of Geographers for exceptional work at the intersection of policy and basic research. He is also the recipient of the 2024 Edward L. Ullman Award of the American Association of Geographers for lifelong contributions to transportation geography. His research has been funded by SSHRC, NSERC, CIHR, and HRSDC, which reflects the breadth of his scientific endeavors. Prof. Páez has published widely, including 166 peer-reviewed papers in international journals and three edited books. He is also co-author of the book *Discrete Choice Analysis with R*. Prof. Páez is among the top 20 transportation researchers in the world whose careers started after 1999, according to Elsevier-Stanford's ranking of top world scientists. He is best known for his social science theory-informed quantitative research. His work on *normative accessibility* and *relative accessibility* is an effort to implement equity ideals in accessibility analysis. Prof. Páez is also well-versed in theoretical and qualitative approaches. He will coordinate the project and provide overall direction, in addition to participating in the supervision of graduate students and conducting research. **This project would be the primary research subject of the PI and he is committed to investing 80% of his research time in it.**

Léa RAVENSBERGEN (co-applicant, relative contribution to project 15%) is an early career researcher of equity in sustainable urban mobility. Since beginning at McMaster in 2023, she has supervised 1 PhD student and 6 MA students, 6 of which has been granted graduated SSHRC Canada Graduate Scholarships. She is a mixed-methods researcher who has led multiple qualitative research projects focused on transportation users' experiences, including two that utilized a Codesign framework. In 2025, she led a series of Codesign focus groups with older adults on how best to design mobility reporting to encourage active lifestyles (currently under review in *Journal of Aging and Physical Activity*). In 2026, she worked with the Hamilton's transit operator Hamilton Street Railway (HSR) to Codesign how best to deliver paratransit as the City of Hamilton moves toward integrated transit. Prof. Ravensbergen will lead the Codesign component of the project, and **she is committed to dedicating 15% of her research time to this project.**

Rafael H.M. PEREIRA (collaborator, relative contribution to project 10%) is Director of Data Science at IPEA, Brazil's federal economics research institute. He is an Oxford-trained geographer who specializes in transport policy and justice, cities and equity, and spatial data science. His contributions to transportation justice are largely responsible for shaping the field (20,28,68,69), while his contributions to numerous publicly available *R* and *Python* packages have been hugely influential in changing the practice of transportation research and planning. Open-source software such as {r5r} (60) and {r5py} (70) (which collectively have over a million downloads) are now routinely used by community planners, geographic researchers, and other members of the public for the purpose of simulating routing by different transportation modes. He has also worked on several packages for working with general transit feed specification data such as {gtfstools} (70). His role will be to provide advice on the theory and practice of transportation justice, and he will also lend his considerable expertise to the development of open-source tools. **His time commitment to the project is 5% of his research time.**

B. DESCRIPTION OF PREVIOUS AND ONGOING RESULTS

Christopher D. HIGGINS has done research with conceptual, methodological, and applied outputs that are relevant to the proposed work. This includes the development of place-based/person-based (71), as well as competitive accessibility measures (39,72) that improve our understanding of who can reach destinations using the transportation network. His applied studies have incorporated equity into route-level transit performance measurement and examined disparities affecting the access of low-income residents, and essential workers to health and social services (48,73,74). Complementary research has assessed relationships among transit accessibility, residential location, and housing prices within individual metropolitan regions and across Canada (75). His ongoing research extends these approaches to the joint analysis of equity in accessibility and housing affordability. Further, Prof. Higgins brings a commitment to open, replicable, and reproducible research as evidenced through the development of the {pynogrid} package (76) and his work on a forthcoming open access book "Advanced Accessibility Analysis with R" with Prof. Páez and Dr. Pereira. These skills mean that EquiTransit is well-grounded

for developing theoretically-informed equity indicators, for conducting comparable analyses across Canadian metropolitan areas, and for communicating complex patterns through accessible analytical tools.

Juanita DE BARROS (co-applicant) main field of research is into human rights and justice, with a focus on the political and social factors that shaped policies dealing with the intersection of health and the law in the past, their impact on the lived experience of women and children especially, and the longer-term impact of these policies. Her research projects include a sole-authored monograph on gender and reproductive health in the Caribbean (77), a forthcoming co-written article on urban vagrancy laws in late nineteenth-century Guyana, and a multidisciplinary co-edited essay collection (with McGill-Queens University Press). This latter project addresses child protection policies and discourse from the nineteenth to the early twenty-first centuries and the varied ways that these policies served adult institutional and political interests, often resulting in the forced labour and incarceration of children. This project is Prof. De Barros first foray into transportation-related research, and we believe that her historian's eye, in addition to her expertise in human rights and justice, will be an asset to the proposed work. Historical perspectives on transportation systems remain rare despite their undeniable value (78,79). The opportunity to develop an ongoing collaboration between historians and transportation scholars is a unique aspect of the project.

Moataz MOHAMED (co-investigator) main field of expertise developing and evaluating indicators of transit service performance and quality from both operator and rider perspectives (80–82) is directly relevant to the project. He has led qualitative and mixed-methods investigations into what current and potential riders desire from transit, including grounded-theory analyses of the barriers travelers face (83,84), and has studied transit users' satisfaction among population groups with distinct mobility needs (85). Prof. Mohamed is an expert in modelling travel behaviour and preference heterogeneity across rider personas and generational cohorts (86–88), as well as in the development of accessibility and spatial-analysis measures for transportation (72,89,90). He has direct experience translating research into open, web-based decision-support tools for public agencies and practitioners, including work applying open-source data and machine learning to inform transportation decision-making (91). He has developed public-facing tools for the Ontario Ministry of Transportation. Currently, Mohamed is Director of the McMaster Institute for Transportation & Logistics and PI/Co-PI on a portfolio of NSERC-, NRCan-, and municipally funded projects on electric and zero-emission transit system planning, resilience, and equitable transit service design.

Antonio PÁEZ has extensive experience related to this proposal. At an early stage he became interested in the transportation challenges faced by older adults (92). Within a few years, this had expanded to research on the travel behavior of populations of special interest, including seniors (93–95), single parents (96,97), and children (98–100). He has a longstanding interest in the implications of social science theory and policy on methodology, as illustrated by his work on normative accessibility (101) and relative accessibility (102). These works, initially motivated by the policy needs of Human Resources and Social Development Canada (an agency of Canada's federal government) were an effort to clarify the distinction between policy goals and travel behavior and to provide benchmarks for policy analysis. His interests have since come to encompass broader issues of equity and justice in transportation (66,73,103). As a member of the leadership group of the SSHRC-funded project Mobilizing Justice, Prof. Páez work was on data-driven standards for equity analysis, which led to novel tools for the analysis of accessibility (39,72,104), clearer theoretical foundations for developing standards of equity (41), and the adoption of dignity as the root for human rights and justice in transportation (105). Furthermore, Prof. Páez is an expert in the development and application of spatial analysis tools with a focus on transportation (89,96,106) and the development of web-based tools for public use (107,108). It is worth noting that he is also fluent in the use of qualitative methods (84,109). His breadth of knowledge and expertise is an asset for this project, in his role as PI and transversal conduit between the various work packages.

Léa RAVENSBERGEN has extensively researched the transit experiences of equity-seeking groups (85,110,111) and the systemic barriers transit agencies face in planning just systems (110,112,113). She is an experienced qualitative methods researcher (111,114–116) including mobile ethnography (117,118). Specific to this proposed research, she has conducted Codesign research in which older adults designed a mobility reporting tool with the research team (119). She also has expertise in community-based research (120) and has worked in collaboration with all levels of governments (e.g., The Public Health Agency of Canada, Transports et mobilité durable Québec, the City of Hamilton), transit agencies across the country (e.g., Edmonton Transit, TTC, York Region Transit), and non-for-profit organizations (e.g., Hamilton Bike Share, Bike Host). She is part of a pan-

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Canadian research team that was granted a \$.16 million NSERC Alliance grant titled "Development of data-driven tools for planning and operating gender-inclusive public transit systems in Canada". In this role, she is working in partnership with 14 transit agencies across Canada to shed light on how to better integrate and implement gender-sensitive data collection practices. Relevant to the proposed research project, since 2023 she has been working with the Hamilton Street Railway (HSR), the public transit agency of the City of Hamilton (110,121,122). She is well positioned to lead to the Codesign element of this research, as she has built the relationships required to ensure a successful process.

C. DESCRIPTION OF TRAINING STRATEGIES

Our strategy for training highly qualified personnel is based on extensive involvement of the students, well-defined responsibilities for the integration of graduate students into all aspects of the project, and close supervision by a team of interdisciplinary supervisors (see Table MT.1).

Table MT.1 Supervisory Matrix

			Year	Main Supervisor	Supervisory Committee
WP.1 Theory & Equity Indicators	Post-Doctoral Fellow 1	PD1	1	Páez	De Barros, Pereira
WP.1 Theory & Equity Indicators	Post-Doctoral Fellow 2	PD2	5	Páez	De Barros, Pereira
WP.2 Codesign	Ph.D. Student 1	PhD2	1 - 4	Ravensbergen	Páez, De Barros
WP.3 Analytics & Tools	Ph.D. Student 2	PhD1	2 - 5	Higgins	Páez, Pereira
WP.4 Equity Informed Planning & Evaluation	Master Student 1	M1	1 - 2	Mohamed	Páez, Higgins
WP.4 Equity Informed Planning & Evaluation	Master Student 2	M2	3 - 4	Mohamed	Páez, Higgins

The focus of PhD1 will be on the codesign and will therefore develop a variety of transferable skills, including soft skills such as communications and time management. PhD2 will focus on geospatial technologies and the development of open-source tools. M1 and M2 will participate in the development of equity-based assessment and evaluation and will learn valuable skills like coding and optimization. PD1 and PD2 will have an opportunity to work across work packages and will learn research management, communication, in addition to the creation of substantive knowledge. Through the project, all trainees will work on publication and communication skills, as they will be expected to publish their findings as lead authors, and will develop a variety of soft and transferable skills (see Table MT.2). Training for the research will be taught in part by members of the research team, through a combination of close supervision and regular discussion, and in part through enrolment in appropriate graduate courses offered at our institutions [e.g., reproducible research; (123)]. As described in the Detailed Description of the project, there are several objectives and methodological approaches to this project. The students will work with the professors to develop their own research questions related to their subject. In our experience, students often bring ideas that enrich the project's original goals and doing so generally increases their motivation.

Table MT.2 SkillsMatrix

		Year	KS	QL	QT	PP	DL	PC	DM	RE	CCE	TW	IIC	TM	BM
Theory & Equity Indicators	PD 1	1													
Theory & Equity Indicators	PD 2	5													
Analytics & Tools	PhD 1	1 - 4													
Equity Informed Planning & Evaluation	M 1	1 - 2													
Equity Informed Planning & Evaluation	M 2	4 - 5													
Codesign	PhD 2	2 - 5													

Skills (Key): KS: Knowledge Synthesis; QL: Qualitative research methods; QT: Quantitative research methods; PP: Publication and presentations (communication skills); DL: Digital literacy; PC: Programming and coding; DM: Data management and analysis; RE: research ethics; CCE: Consultation and community engagement; TW: Teamwork; IIC: International and intersectoral conference; TM: Time management; BM: Budget management.

Budget Justification (max. 2 pages)

Personnel costs (\$464,000 = 92.8%)

The largest portion of the costs are allocated towards the students (master's and PhD) and post-doctoral research fellows. Stipends for master's students are \$25,000 per year (years 1-2; 3-4), stipends of PhD students are \$28,000 per year (years 1-4; 2-5), and the Post-Doctoral Fellows will be paid \$70,000 per year (years 1 & 5). Over the duration of this 5-year project, this will account for 92.8% of the funds requested. A timeline of when master's students, PhD students, and post-doctoral fellows will be assisting in the project can be seen in Table MT.1.

Travel costs (\$24,000 = 4.8%)

Beginning in year 2 of the project, \$2,000 will be allocated to students per year (in total, hence ~\$600 to \$1,000 per student per year) for domestic travel to Canadian conferences such as the Canadian Associations of Geographers Annual Conference, and other more transportation-oriented conferences such as the Canadian Transportation Research Forum. Additionally, \$3,000 will be allocated to students for international travel to various conferences in years 2 and 3 of the project (\$1,000 to \$1,500 per student per year); as well as \$5,000 allocated to students for international travel in years 4 and 5 (~\$1,600 to \$2,500 per student per year). These funds will cover conference registration fees, transportation to the conference, and hotel stays. Regardless of year of study, students will be encouraged to attend the conferences as well as present in their upper years of study, as it is crucial for not only showcasing the research project, but also learning from other experts in the fields of transportation, geography, and planning. Hence, 4.8% of the total project cost will be allocated to travel costs.

Computer hardware (\$4,750 = 0.95%)

The quantitative research segments of this project, namely the geospatial analysis, the development of the equity-informed evaluation tool, and the development of open-source packages, will require specialized equipment. This translates into computers with sufficient processors, RAM, and storage to be able to complete various computing processes and generating data. Members of the research team have some computers in their equipment inventory, and these will be used for this project. Funds are requested to upgrade or replace ageing equipment on years 2 and 3 of the project. We are budgeting \$2,500 in year 2 of the project, and \$2,250 in year 3 of the project to this end. These funds will be allocated to hardware updates, computer repairs, and the acquisition of various peripheral equipment as needed. Together with computer equipment already owned by members of the research team, the aim is to maintain four fully equipped workstations for the duration of the project, with one workstation assigned to each of the four work packages.

Other expenses: Codesign (\$7,250 = 1.45%)

The qualitative research component of the project is the CODESIGN, with an estimated budget of \$6,000 to provide incentives for participants across different focus groups (based on factors such as socioeconomic status and/or neighbourhood).

Each codesign focus group participant will receive a \$50 gift card and their transport to and from McMaster reimbursed (\$4.00 x 2 (return bus fare) = \$8.00). All co-design focus group materials (e.g., printing, Bristol boards, etc.) will be provided by Dr. Ravensbergen's lab in kind. There will be two codesign phases: development of indicators and development of tools.

Development of indicators

Year 2: \$2,610

Codesign Step 1 (Empathy) (\$1,160.00): series of approximate 4 focus groups with 5 transit user participants in each (4 x 5 participants x \$58.00 per participant = \$1,150.00)

Steps 2 (Define) and 3 (Ideate) focus groups (\$1,450.00): 3 focus groups with 5 transit user participants (3 x 5 participants x \$58.00 per participant = \$870) and 2 focus groups with 5 transit providers (2 x 5 participants x \$58.00 per participant = \$580.00)

Year 3: \$1,160.00

Step 5 (Test) (\$1,160.00): final 3 focus groups with 4 transit user participants (3 x 4 participants x \$58.00 per participant = \$696.00) and 2 focus groups with 4 transit providers (2 x 4 participants x \$58.00 per participant = \$464.00)

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Development of tools

Year 4: \$2,320

Codesign Step 1 (Empathy) (\$1,160.00): series of approximate 4 focus groups with 4 transit user participants in each (4 x 4 participants x \$58.00 per participant = \$928.00)

Steps 2 (Define) and 3 (Ideate) focus groups (\$1,160.00): 3 focus groups with 4 transit user participants (3 x 4 participants x \$58.00 per participant = \$696) and 2 focus groups with 4 transit providers (2 x 4 participants x \$58.00 per participant = \$464.00)

Year 5: \$1,160.00

Step 5 (Test) (\$1,160.00): final 3 focus groups with 4 transit user participants (3 x 4 participants x \$58.00 per participant = \$696.00) and 2 focus groups with 4 transit providers (2 x 4 participants x \$58.00 per participant = \$464.00)

Software (\$0)

This project will use exclusively open-source software to reduce the cost of research and the cost of use for other users, and for reproducibility purposes.

Open access publishing (\$0)

We will leverage publishing agreements made by our institutions with reputable publishers such as Wiley, PLoS ONE, Elsevier, and Springer.

Hence, the total amount requested for the five-year project is **\$500,000**.

Expected outcomes (max. 3,800 characters)

The project assembles a group of researchers with a proven record of collaboration, and it fosters new collaborations between several research professors. This includes new collaborative opportunities between Ravensbergen and Higgins, Mohamed and Pereira, Ravensbergen and Pereira, and all transportation scholars collaborating with De Barros, an expert in human rights and justice. The team takes previous collaborations and expands them in novel ways that will benefit the quality of the scholarship produced. Moreover, the codesign framework proposed builds on existing collaborations (e.g., between Co-PI Ravensbergen and Mohammed and Hamilton's transit agency) and fosters new connections with community stakeholders. The project aspires to empower users of transit to have a more fluid participation in decision-making processes that affect them, and to help providers of a key public service to better incorporate user knowledge. In this way, the codesign framework opens avenues for knowledge exchange between providers and users of transit and thus contributes to enhance the process and practice of effective governance.

Through this project, four graduate students and two post-doctoral researchers will be trained and gain various new skills, to the betterment of Canada's transportation and research expertise. Other outcomes of the project are of three kinds: communication, research materials, and tools. The project will produce ten peer-reviewed papers (1 article by each Masters student, 3 articles per Doctoral student, and 1 article by each post-doctoral researcher). In addition, there will be communications for lay audiences, in the form of case studies, policy briefs, and user's manuals. New data produced during the research will be shared in the form of open data products, which will be made available via web repositories in addition to registering them for digital object identification (doi) using Zenodo. Other research materials, such as research documents and computational notebooks, will be shared using public web repositories. The project aims to push the boundaries of current practices in open science by expanding on the current expertise of members of the research team. The project will also produce open-source tools (e.g., dashboards) to help stakeholders assess the equity implications of various transit provision scenarios. More generally, knowledge transfer will prepare the next generation of transportation equity champions.

With better knowledge, better choices will be facilitated for planners and policymakers at transit agencies and users of public transportation. Rather than basing decisions on relatively narrow fiscal considerations, transparent and publicly available tools will contribute to advance more equitable public transportation planning. As a matter of procedural justice in transportation, planning must consider all segments of society, and for participation to be meaningful relevant information must be accessible. The tools proposed contribute to fill an information gap about levels of service by clarifying in a more holistic way the distribution of transit's costs and benefits. The development of visual tools based on research that accounts for relevant perspectives of stakeholders will enhance public discourse, public policy, and planning. Generating knowledge about the equity implications of transit operations will help steer conversations from a predominantly economic perspective to a fuller accounting of transit's numerous non-monetary benefits. Generating the tools to implement this knowledge will help inform policy and practice related to urban planning, transport, social inclusion, and community development by translating abstract research into trustworthy, locally relevant, and actionable information.

Research Contributions: Antonio Páez

Refereed contributions BOOKS 2020-2026 (*: SSHRC supported; +: Student author)

- * 1. Páez, A. & Boisjoly, G. (2026). *Discrete Choice Analysis with R*. Springer Cham.
<https://doi.org/10.1007/978-3-031-20719-8>

Refereed contributions JOURNAL ARTICLES 2020-2026 (*: SSHRC supported; +: Student author) (54 peer-reviewed journal publications in this period, only 36 most relevant to proposal are listed below)

- * 1. dos Santos, B. D. (+), Moghadasi, M. (+), & Páez, A. (2026). A 30-year analysis of active travel behaviour in Canada. *Journal of Transport Geography*, 131, 104561.
<https://doi.org/10.1016/j.jtrangeo.2026.104561>
- * 2. Higgins, C. D., Li, C., & Páez, A. (2026). Person-based aggregate space-time accessibility (PASTA): Bridging the gap between place- and person-based accessibility. *Journal of Transport Geography*, 132, 104602. <https://doi.org/10.1016/j.jtrangeo.2026.104602>
- * 3. Soukhov, A. (+) Pereira, R. H. M., Higgins, C. D., & Páez, A. (2025). A family of accessibility measures derived from spatial interaction principles. *PLOS ONE*, 20(11), 1–31.
<https://doi.org/10.1371/journal.pone.0335951>
- * 4. dos Santos, B. D. (+), Moghadasi, M. (+), & Páez, A. (2025). ActiveCA: Time use data from the general social survey of Canada to study active travel. *Environment and Planning B: Urban Analytics and City Science*, 52(8), 2037–2047. <https://doi.org/10.1177/23998083251374724>
- * 5. Soukhov, A., Ravensbergen, L., Dorantes, L. M., & Páez, A. (2025). Towards completely caring 15-minute neighbourhoods. *Transportation Research Part A: Policy and Practice*, 198, 104503.
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Forthcoming contributions JOURNAL ARTICLES (*: SSHRC supported; +: Student author)

- * 1. Soukhov, A. (+), Tiznado-Aitken, I., Palm, M., Farber, S., & **Paez, A.** (2025). Standards of fairness in the transportation literature. *Transportation*. <https://doi.org/10.1007/s11116-025-10693-8> (in press)
- * 2. Soukhov, A. (+), Higgins, C. D., Mohamed, M., & **Páez, A.** (2025). Ten years of school closures and consolidations in Hamilton, Canada and the impact on multimodal accessibility. *Spatial Networks and Economics*. <https://doi.org/10.1007/s11067-025-09677-z> (in press)
3. Dos Santos, B.D. (+), Hansen, T., & **Páez, A.** (2026). Transportation and dignity: A scoping review and theoretical framework. *Transport Reviews* (revised and submitted)
- * 4. Jamal, S., & **Páez, A.** (2026). Changes in walking frequency and its equity implications during the first wave of COVID-19 in Bangladesh. *Journal of Transport & Health* (revised and submitted)
- * 5. Dos Santos, B.D. (+), Soukhov, A., & **Páez, A.** (2026) Equity Standards for transport policy: using objective accessibility, perceived accessibility and employment outcomes to set thresholds for equity planning. *Transportation Research Part A: Policy and Practice* (submitted)

Other research contributions (2020-2026)

- In the period 2020-2026 I was co-author of two reports for the Mobilizing Justice project and two book chapters.
- Editor-in-Chief, *Journal of Geographical Systems*.
- Member of the Leadership Committee, Mobilizing Justice.
- Member of the International Scientific Committee, Chaire Mobilité, Polytechnique Montreal.
- Guest Editor, Special Issue on Elderly's Mobility and the Living Environment. *Journal of Transport Geography*.
- Editorial Board Member (*Annals of the Association of American Geographers*; *Economic and Business Letters*; *Geographical Analysis*; *International Journal of Geographical Information Science*; *Papers in Regional Science*; *Transportation*).
- 36 invited talks (17 in Canada, 6 in Spain, 4 in Brazil, 3 in USA, 2 in Australia, 2 in UK, 1 in Italy, 1 in Portugal).

Most significant career research contributions

A total of 177 publications in my record are indexed in Scopus. These publications have been cited 8,473 times (as of September 23, 2026). The average number of citations per indexed publication is 47.87, and the h-index is 50 (50 publications have been cited 50 times or more). Most significant career contributions are:

1. **Páez A**, Scott DM, Morency C (2012). Measuring accessibility: positive and normative implementations of various accessibility indicators, *Journal of Transport Geography*, 25 141-153. 141–153. <https://doi.org/10.1016/j.jtrangeo.2012.03.016> (**656 citations**). This research was key to distinguishing between behavioral processes and policy aims in accessibility analysis. A key finding was the identification of critical issues in the way accessibility measures are interpreted depending on their implementation. By

providing concrete guidance for interpretation, this paper has become a key reference in accessibility research and has been cited in the fields of transportation, geography, economics, environmental studies, civil engineering, urban studies, and operations research.

2. **Páez, A., & Whalen, K. (+)** (2010). Enjoyment of commute: A comparison of different transportation modes. *Transportation Research Part A-Policy and Practice*, 44(7), 537–549. <https://doi.org/10.1016/j.tra.2010.04.003> (**273 citations**). This paper provided evidence of the intrinsic utility of active travel and challenged the conventional wisdom that travel time is always a disutility, with important implications for policy. Kate Whale was a Master student when this paper was written. It has become a valued reference for research in the fields of transportation, geography, business economics, public administration, and engineering.
3. **Páez A, Mercado RG (+), Farber S. (+), Morency C, Roorda M** (2010). Relative Accessibility Deprivation Indicators for Urban Settings: Definitions and Application to Food Deserts in Montreal, *Urban Studies*, 47 (7) 1415-1438. <https://doi.org/10.1177/0042098009353626> (**234 citations**). The methods presented in this paper were an early attempt at obtaining benchmarks for equity analysis. They provided the tools to assess differences in accessibility from a selected baseline, similar to the way poverty lines are measured with respect to some judiciously selected “median household”. This paper has become an important resource for research in geography, transportation, economics, environmental studies, public health, and urban studies.
4. **Reyes, M.(+), Páez, A., & Morency, C.** (2014). Walking accessibility to urban parks by children: A case study of Montreal. *Landscape and Urban Planning*, 125, 38–47. <https://doi.org/10.1016/j.landurbplan.2014.02.002> (**196 citations**). This research was led by Mario Reyes as a Master student, and it was among the first papers to examine the provision of green spaces from the perspective of children. This paper has been cited in the fields of environmental sciences, economics, public health, geography, and public administration.
5. **Black J, Páez A, Suthanaya P, (+)** (2002), Sustainable Urban Transportation: Performance Indicators and Some Analytical Approaches, *ASCE Journal of Urban Planning and Development*, 128 (4) 184-209 (**195 citations**). Sustainability has long been an interest of mine, particularly the social pillar of sustainability. This collaboration with John Black (University of New South Wales) and Putu Suthanaya (who at the time was a doctoral candidate) has been influential in civil engineering, transportation, urban studies, economics, and environmental science.

Contributions to training

Since 2020, I have supervised 8 senior undergraduate students in the programs of Engineering and Society (3), Geography (2), Life Sciences (2), and Economics (1). All these students completed a capstone project or thesis. I supervised 4 master’s students (3 in Geography, 1 in Public Health) and cosupervised 3 master students (2 in Geography and 1 in eHealth). I supervised 2 doctoral students (both in Geography, including a recipient of a Canada Graduate Scholarship award) and cosupervised 1 doctoral student in Civil Engineering. I currently supervise two master’s students in Geography and two doctoral students in Geography (including 1 Vanier Scholar). One master student (Desjardins) was the winner of the Master’s Thesis Competition in Transport Geography by the American Association of Geographers (2021). One doctoral student was the winner of the Doctoral Thesis Competition in Transport Geography by the Association of American Geographers (2025). Graduate students working under my supervision are always offered opportunities to participate in research. Graduate students have participated in 37 out of a total of 54 peer-reviewed publications on record in the period since 2020, and their contributions have been duly recognized with coauthorships. The contributions of one undergraduate student working under my supervision were recognized with coauthorship in one peer-reviewed publication in 2024. Students that have trained under my supervision have moved on to academic careers (e.g., University of Toronto, Xi’an Jiaotong University), to careers in government (e.g., Statistics Canada), the broader public sector (Canada Housing and Mortgage Corporation), and the private sector.

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